
Machine Learning Based Analog Circuit Fault Diagnosis
PDF 14 — Chapter 21: Complete Viva Preparation
50 Core Viva Questions | 60+ Advanced Professor Cross-Questioning Scenarios

PART 1 — The 50 Most Important Viva Questions

Chapter 21 — Complete Viva Preparation Part 1 — The 50 Most Important Viva Questions (With Perfect Answers) Before Starting Congratulations. You have completed the entire project. Now comes the final challenge. The Viva. Most students lose marks not because they don't know the project, but because they cannot explain it clearly. This chapter will prepare you for that. Rule Never memorize answers. Understand them. Then answer in your own words.

Section 1 — Basic Questions Q1

Tell me about your project.

Perfect Answer

Our project focuses on analog circuit fault diagnosis using machine learning. We generated circuit responses using LTspice, extracted 17 meaningful features from the transient waveforms, trained machine learning models such as SVM and ANN for fault classification, and developed a dashboard that allows users to upload waveform data and automatically identify the corresponding circuit fault.

Q2

What problem does your project solve?

Perfect Answer

The project automates fault identification in analog circuits. Instead of manually analyzing waveforms, machine learning classifies faults quickly and consistently using extracted signal features.

Q3

Why did you choose this topic?

Perfect Answer

Analog fault diagnosis is traditionally time-consuming and requires expert knowledge. We wanted to investigate whether machine learning could automate the process while maintaining high accuracy.

Q4

Why analog circuits?

Perfect Answer

Analog circuits are widely used in communication systems, medical devices, industrial electronics, and power systems. Fault diagnosis in such circuits is an important engineering problem.

Q5

Why Machine Learning?

Perfect Answer

Machine learning can learn relationships between extracted signal features and fault classes, allowing automatic diagnosis without manually defining decision rules.

Section 2 — Dataset Questions Q6

How was the dataset created?

Perfect Answer

We simulated different fault conditions using LTspice. The generated waveforms were exported, processed, and converted into feature vectors for machine learning.

Q7

Why LTspice?

Perfect Answer

LTspice provides accurate analog circuit simulation, making it possible to generate large datasets under controlled conditions without requiring expensive laboratory hardware.

Q8

How many features were extracted?

Perfect Answer

We extracted 17 features representing different characteristics of the waveform, including statistical, transient, energy-based, and steady-state information.

Q9

Why feature extraction?

Perfect Answer

Raw waveforms contain thousands of samples. Feature extraction summarizes the important characteristics into a compact numerical representation suitable for machine learning.

Q10

Why 17 features?

Perfect Answer

Initially we used fewer features. During confusion matrix analysis, we identified missing steady-state information and introduced DC Level as the seventeenth feature to improve fault discrimination.

Section 3 — Machine Learning Questions Q11

Why SVM?

Perfect Answer

SVM performs well on small and medium-sized datasets, handles nonlinear relationships using the RBF kernel, and provides strong classification performance with relatively few hyperparameters.

Q12

Why ANN?

Perfect Answer

ANN was implemented to compare a deep learning approach with SVM and determine which model was more suitable for our dataset.

Q13

Which model performed better?

Perfect Answer

SVM was more suitable for our dataset because of its size, carefully engineered features, and lower model complexity.

Q14

Why RBF kernel?

Perfect Answer

The extracted feature vectors were not perfectly linearly separable. The RBF kernel allowed the SVM to learn nonlinear decision boundaries.

Q15

Why StandardScaler?

Perfect Answer

SVM is sensitive to feature scales because it relies on distance calculations. StandardScaler ensures all features contribute fairly during training.

Section 4 — GridSearchCV Questions Q16

Why GridSearchCV?

Perfect Answer

GridSearchCV systematically evaluated multiple combinations of hyperparameters using cross-validation and selected the best-performing model automatically.

Q17

What is Cross Validation?

Perfect Answer

Cross Validation evaluates the model using multiple train-test splits and averages the results to obtain a more reliable estimate of performance.

Q18

Why not manually choose C and Gamma?

Perfect Answer

Manual selection is subjective and may not produce the optimal model. GridSearchCV provides an objective and systematic approach.

Section 5 — Evaluation Questions Q19

How did you evaluate the model?

Perfect Answer

We evaluated the classifier using Accuracy, Precision, Recall, F1 Score, Classification Report, and Confusion Matrix.

Q20

Why isn't Accuracy enough?

Perfect Answer

Accuracy only provides overall performance. The Confusion Matrix reveals which fault classes are being confused.

Q21

What is Precision?

Perfect Answer

Precision measures how many predicted positive samples are actually positive.

Q22

What is Recall?

Perfect Answer

Recall measures how many actual positive samples are correctly detected.

Q23

What is F1 Score?

Perfect Answer

F1 Score is the harmonic mean of Precision and Recall and provides a balanced evaluation metric.

Section 6 — Engineering Questions Q24

How did you improve the model?

Perfect Answer

We analyzed the confusion matrix, identified recurring RC_Open misclassification, investigated the extracted features, introduced DC Level, and later incorporated the Topology Rule.

Q25

Why was DC Level introduced?

Perfect Answer

Existing features did not explicitly capture steady-state voltage. DC Level represented this missing information and improved classification.

Q26

What is the Topology Rule?

Perfect Answer

The Topology Rule combines machine learning predictions with engineering knowledge about circuit behavior to improve diagnostic reliability.

Q27

Why didn't you simply change the classifier?

Perfect Answer

The problem was caused by missing information rather than classifier limitations. Improving the feature representation was a more meaningful solution.

Section 7 — Dashboard Questions Q28

What is the purpose of the dashboard?

Perfect Answer

The dashboard provides an easy-to-use interface for uploading waveform data, extracting features, running the trained model, and displaying fault predictions.

Q29

Does the dashboard train the model?

Perfect Answer

No. Training is performed offline. The dashboard only loads the trained model and performs prediction.

Q30

What happens after the user uploads a file?

Perfect Answer

The dashboard validates the file, extracts the features, applies StandardScaler, loads the trained model, predicts the fault, and displays the results.

Section 8 — Limitations Q31

What are the limitations?

Perfect Answer

The project currently uses simulated LTspice data, supports only selected circuit topologies and fault classes, performs offline prediction, and has not yet been validated using physical hardware.

Q32

How would you improve it?

Perfect Answer

I would validate the model using real circuits, implement real-time monitoring, integrate IoT, deploy the model on edge devices, and expand the dataset.

Section 9 — Difficult Questions Q33

Why not Deep Learning?

Perfect Answer

Deep learning generally requires much larger datasets. For our moderate-sized dataset with carefully engineered features, SVM provided a better balance between performance and complexity.

Q34

Can your model classify unknown faults?

Perfect Answer

No. The classifier can only recognize fault classes included in its training dataset.

Q35

Can this be used in industry?

Perfect Answer

Yes, after hardware validation, real-time data acquisition, and integration with industrial monitoring systems.

Q36

Why not Random Forest?

Perfect Answer

Random Forest is a strong classifier, but SVM is particularly effective for moderate-sized datasets with nonlinear feature spaces. We focused on SVM because it matched the characteristics of our problem.

Q37

Why not CNN?

Perfect Answer

CNNs are more suitable when using raw waveform or image-like inputs. Our project used engineered numerical features, making SVM a more appropriate choice.

Q38

What is the novelty of your project?

Perfect Answer

Beyond standard machine learning, the project introduced iterative engineering improvements through confusion matrix analysis, the addition of the DC Level feature, and the Topology Rule, combining machine learning with circuit-domain knowledge.

Section 10 — Final Questions Q39

What did you personally learn?

Perfect Answer

I learned that successful machine learning projects require not only selecting algorithms but also understanding the underlying engineering problem, designing meaningful features, evaluating errors, and iteratively improving the system.

Q40

What is your biggest contribution?

Perfect Answer

The biggest contribution was improving the fault diagnosis pipeline by systematically analyzing classifier errors and incorporating additional engineering knowledge through feature engineering and the Topology Rule.

MEMORY TRICK

Golden Viva Tip If you don't know an answer, never say "I don't know." Instead say "I haven't implemented that in this project, but based on my understanding." This shows confidence and technical maturity. Common Viva Mistakes [X] Giving one-word answers. [X] Memorizing definitions without understanding. [X] Saying "because that's what the code does." [X] Guessing when unsure. [X] Interrupting the examiner.

REVISION NOTES

Remember the viva structure. Problem Dataset Features SVM ANN GridSearchCV Evaluation RC_Open investigation DC Level Topology Rule Dashboard Limitations Future Scope If you can explain these 12 topics clearly, you can answer the majority of viva questions confidently.

End of Chapter 21 (Part 1) Congratulations. You now have a strong foundation for your viva preparation.

Author's Note The next continuation will contain 60+ advanced professor-level cross-questioning scenarios. These are the follow-up questions professors ask after your first answer, designed to test deep understanding rather than memorization. Mastering those will prepare you for even the toughest project defenses.

PART 2 — Advanced Professor Cross-Questioning

Chapter 21 — Complete Viva Preparation Part 2 — Advanced Professor Cross-Questioning (60+ Follow-up Questions) Before Starting Now we move to the hardest part of the viva. These are NOT direct questions. These are the questions professors ask after you answer the first question. This is called cross-questioning. Most students fail here. After this chapter, you won't.

Cross Question 1

PROFESSOR ASKS

Why didn't you directly use the raw waveform instead of extracting features?

EXCELLENT ANSWER

Raw waveforms contain thousands of sample points, making training computationally expensive and increasing the dimensionality of the problem. Feature extraction summarizes the important characteristics of the waveform into a compact numerical representation that is easier for machine learning algorithms like SVM to process.

PROFESSOR ASKS

Can deep learning use raw waveforms?

EXCELLENT ANSWER

Yes. Models such as CNNs and LSTMs can learn directly from raw waveforms. However, they generally require significantly larger datasets and higher computational resources than were available for this project.

Cross Question 2

PROFESSOR ASKS

Why exactly 17 features?

EXCELLENT ANSWER

Initially, the feature set contained fewer features. During confusion matrix analysis, we observed recurring misclassification of RC_Open. Investigation showed that steady-state voltage was not explicitly represented, so DC Level was added as the seventeenth feature.

PROFESSOR ASKS

Could there be an 18th feature?

EXCELLENT ANSWER

Yes. Feature engineering is an iterative process. Additional features such as spectral entropy, wavelet coefficients, or higher-order statistical measures could further improve performance if supported by experimental evaluation.

Cross Question 3

PROFESSOR ASKS

How do you know Peak is important?

EXCELLENT ANSWER

Peak amplitude reflects the maximum response of the transient waveform. Different faults modify the circuit response differently, making Peak a useful discriminating feature. Its usefulness was confirmed experimentally during classifier training and evaluation.

PROFESSOR ASKS

What if Peak is removed?

EXCELLENT ANSWER

The classifier would lose one source of information. Whether performance decreases depends on how much unique information Peak contributes relative to the remaining features. This can be verified experimentally using feature ablation studies.

Cross Question 4

PROFESSOR ASKS

Why not use only DC Level?

EXCELLENT ANSWER

DC Level captures only the steady-state behavior. It provides no information about transient characteristics such as rise time, overshoot, energy, or oscillations. Therefore, it must be combined with other features.

PROFESSOR ASKS

Why not use only RMS?

EXCELLENT ANSWER

RMS measures signal magnitude but ignores many transient properties. No single feature completely describes the waveform, which is why multiple complementary features are required.

Cross Question 5

PROFESSOR ASKS

How do you know the model isn't memorizing the data?

EXCELLENT ANSWER

We separated the dataset into training and testing sets. The model was evaluated using previously unseen test samples. In addition, GridSearchCV and cross-validation helped improve generalization.

PROFESSOR ASKS

What is memorization called in Machine Learning?

EXCELLENT ANSWER

Overfitting.

Cross Question 6

PROFESSOR ASKS

Suppose I double the dataset. Will accuracy always increase?

EXCELLENT ANSWER

Not necessarily. If the additional data is representative and correctly labeled, performance may improve. However, poor-quality or noisy data can reduce accuracy.

Cross Question 7

PROFESSOR ASKS

Why does StandardScaler improve SVM?

EXCELLENT ANSWER

SVM relies on distance calculations. Without scaling, features with larger numerical values dominate the optimization process. StandardScaler ensures all features contribute on a comparable scale.

PROFESSOR ASKS

Would Decision Trees require StandardScaler?

EXCELLENT ANSWER

Not necessarily. Decision Trees split based on feature thresholds and are generally insensitive to feature scaling.

Cross Question 8

PROFESSOR ASKS

Suppose I replace SVM with KNN. What changes?

EXCELLENT ANSWER

The preprocessing pipeline remains largely unchanged, including feature extraction and scaling. Only the classifier changes. KNN would classify samples based on the nearest feature vectors rather than learning a decision boundary.

PROFESSOR ASKS

Would KNN need scaling?

EXCELLENT ANSWER

Yes. Like SVM, KNN relies on distance calculations, making feature scaling important.

Cross Question 9

PROFESSOR ASKS

Why did RC_Open cause problems?

EXCELLENT ANSWER

RC_Open produced feature vectors that overlapped with other fault classes. The confusion matrix revealed recurring misclassification, leading us to investigate the feature representation.

PROFESSOR ASKS

How did you solve it?

EXCELLENT ANSWER

We introduced the DC Level feature to explicitly capture steady-state voltage and later incorporated the Topology Rule to combine machine learning with engineering knowledge.

Cross Question 10

PROFESSOR ASKS

Suppose your accuracy is 99%. Should we trust the model completely?

EXCELLENT ANSWER

No. High accuracy alone is insufficient. We must also examine the confusion matrix, precision, recall, F1 Score, and verify that predictions are physically consistent.

PROFESSOR ASKS

Why?

EXCELLENT ANSWER

A model can achieve high overall accuracy while still performing poorly on specific fault classes. Therefore, class-wise evaluation is essential.

Cross Question 11

PROFESSOR ASKS

What happens if one feature contains random noise?

EXCELLENT ANSWER

Noise can reduce classification performance by making feature vectors less representative of the actual circuit behavior. Feature selection and preprocessing techniques can help reduce this effect.

Cross Question 12

PROFESSOR ASKS

Why compare ANN if SVM was already working?

EXCELLENT ANSWER

Comparing multiple classifiers provides scientific evidence that the chosen model is appropriate. It demonstrates that the selection was based on experimental evaluation rather than assumption.

Cross Question 13

PROFESSOR ASKS
Why didn't ANN outperform SVM?
EXCELLENT ANSWER

Our dataset was moderate in size and already represented by carefully engineered features. Under these conditions, the RBF-kernel SVM was better suited than a more complex neural network.

PROFESSOR ASKS
Would ANN improve with more data?
EXCELLENT ANSWER

Yes. As the dataset grows substantially, ANN can often learn more complex relationships and may outperform traditional classifiers.

Cross Question 14

PROFESSOR ASKS
What is the biggest engineering lesson from your project?
EXCELLENT ANSWER

The biggest lesson is that improving features and understanding the physical system can be more valuable than simply changing machine learning algorithms.

Cross Question 15

PROFESSOR ASKS
If I give you one year, what will you build?
EXCELLENT ANSWER

I would build a real-time analog fault diagnosis system using physical circuits, sensors, IoT-enabled data acquisition, edge deployment on embedded hardware, and cloud-based monitoring with predictive maintenance capabilities.

□ Five "Danger Questions"

These are questions that often surprise students.

Danger Question 1

Why didn't you use MATLAB? Answer LTspice provided efficient analog circuit simulation, while Python offered a flexible ecosystem for machine learning, data processing, and dashboard development.

Danger Question 2

Can your model explain why it predicted a fault? Answer The current model primarily provides the predicted class. Future work could incorporate Explainable AI techniques such as SHAP or LIME to explain feature contributions.

Danger Question 3

Can this project work for digital circuits? Answer Not directly. Digital circuits have different signal characteristics and fault mechanisms. They would require a different dataset, feature extraction strategy, and retraining.

Danger Question 4

What is the weakest part of your project? Answer The biggest limitation is the reliance on simulated LTspice data. Validation using real hardware would be the most important next step.

Danger Question 5

What are you most proud of? Answer The systematic engineering approach—using confusion matrix analysis to identify weaknesses, improving the feature set with DC Level, and combining machine learning with engineering knowledge through the Topology Rule.

MEMORY TRICK

Golden Rule for Cross Questioning Never answer immediately. Think for 2–3 seconds. Then answer slowly and confidently. It creates a much stronger impression than speaking quickly.

COMMON MISTAKES

MISTAKE: [X] Arguing with the examiner.

MISTAKE: [X] Saying "Google says."

MISTAKE: [X] Changing your answer halfway through.

MISTAKE: [X] Giving unnecessarily long explanations.

MISTAKE: [X] Panicking if you don't know an answer. Ultimate Viva Formula Whenever you're asked a technical question, structure your answer like this: **Definition** — Why it is needed. **How we used it** — Result in our project. Example Question: Why StandardScaler? Answer: Definition: StandardScaler normalizes feature values. Why needed: SVM is sensitive to feature scales. How we used it: We fit it on the training set and transformed both training and test data. Result: It ensured fair contribution from all 17 features and improved model training. This structure works for almost every technical question.

End of Chapter 21 (Part 2) Congratulations. You are now prepared for advanced viva cross-questioning. These are the kinds of follow-up questions that distinguish someone who used a project from someone who deeply understands it.

Author's Note The next chapter will move beyond the academic viva and focus on placement interviews. We'll cover role-specific questions for: SDE interviews Data Analyst roles (ZS, Mu Sigma, etc.) ML/AI Engineer roles Core Electronics interviews We'll also discuss how to present the same project differently depending on the role you're interviewing for, which can significantly improve its impact.